

approach. The probability-based school uses “safe” in a historical context. The probability-based approach is more closely associated with the traditional concepts of wealth accumulation and investment management.

Probability-based models tend to focus on this concept of safe withdrawal rates, a concept that is rather foreign to the safety-first school of thought. Safe withdrawal rates are about systematic withdrawals from a volatile portfolio. The question is: how much can retirees withdraw from their savings, which are invested in a diversified portfolio, while still maintaining sufficient confidence that they can safely continue spending without running out of wealth?

In the early 1990s, William Bengen read misguided claims in the popular press that average portfolio returns could guide the calculation of sustainable retirement withdrawal rates. If stocks average 7 percent after inflation, then plugging a 7 percent return into a spreadsheet suggests that retirees could withdraw 7 percent each year without ever dipping into their principal. Bengen recognized the naïveté of ignoring the real-world volatility experienced around that 7 percent return, and he sought to determine what would have worked historically for hypothetical retirees at different points. He used data extending back to 1926 for US financial markets for his research, which introduced the concept of “sequence-of-returns risk” to the financial planning profession.

The problem he set up is simple: a new retiree makes plans for withdrawing some inflation-adjusted amount from his or her savings at the end of each year for a thirty-year retirement period. For a sixty-five-year-old, this leads to a maximum planning age of ninety-five, which Bengen felt was reasonably conservative. What is the highest withdrawal amount as a percentage of retirement date assets that, with inflation adjustments, will be sustainable for the full thirty years? He looked at rolling thirty-year periods from history (1926 to 1955, 1927 to 1956, etc.). He found that with a 50/50 asset allocation to stocks and bonds (the S&P 500 and intermediate-term government bonds), the worst-case scenario experienced in US history was for a hypothetical 1966 retiree who could have withdrawn 4.15 percent at most. Thus was born what is known as the “4 percent rule.”